Question 1: Configure infrastructure and automate application configuration using Ansible by writing playbooks and applying them to target nodes.

Ansible was used with two target types: the control machine (local connection) and a Docker Ubuntu container acting as a managed node (docker connection plugin). The playbook installed nginx, deployed the application page and verified it, and a second run proved idempotency.

Inventory (ansible/inventory.ini)

```
[local]
localhost ansible_connection=local

[dockernodes]
lab3node ansible_connection=docker ansible_python_interpreter=/usr/bin/python3
```

Playbook (ansible/playbook.yml)

```
---
# Lab 3 - automate infrastructure and application configuration
- name: Configure the control machine (localhost)
  hosts: local
  become: false
  tasks:
    - name: Create a lab report directory
      file:
        path: /tmp/lab3-ansible
        state: directory
        mode: "0755"

    - name: Record the managed hosts
      copy:
        content: "configured by ansible on {{ inventory_hostname }}\n"
        dest: /tmp/lab3-ansible/{{ inventory_hostname }}.txt

- name: Configure the managed node and deploy the app page
  hosts: dockernodes
  become: false
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
        update_cache: true

    - name: Deploy the application index page
      copy:
        content: |
          <h1>Lab 3 - configured by Ansible</h1>
          <p>Host: {{ inventory_hostname }}</p>
          <p>App version: 3.0.0</p>
        dest: /var/www/html/index.html
        mode: "0644"

    - name: Start nginx (container has no systemd)
      shell: "service nginx start || service nginx reload"
      changed_when: false

    - name: Check nginx responds
      uri:
        url: http://localhost/
        return_content: true
      register: nginx_check

    - name: Show the deployed page
      debug:
        msg: "nginx responded HTTP {{ nginx_check.status }} with the deployed page"
```

Applying the playbook

```
$ ansible-playbook -i ansible/inventory.ini ansible/playbook.yml
PLAY [Configure the managed node and deploy the app page] ***
TASK [Install nginx] ... ok
TASK [Deploy the application index page] ... changed
TASK [Start nginx] ... ok
TASK [Show the deployed page] ... ok
PLAY RECAP
lab3node   : ok=6  changed=0  unreachable=0  failed=0
localhost  : ok=3  changed=0  unreachable=0  failed=0
```

```
$ curl http://localhost:8081/   (deployed page on the managed node)
<h1>Lab 3 - configured by Ansible</h1>
<p>Host: lab3node</p>
<p>App version: 3.0.0</p>
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ docker run -d --name lab3node -p 8081:80 ubuntu:24.04 sleep infinity
abhinav@LAPTOP-BSV7PCJK:~$ docker exec lab3node apt-get install -y python3 nginx   # base for the managed
abhinav@LAPTOP-BSV7PCJK:~$ ansible-playbook -i ansible/inventory.ini ansible/playbook.yml
PLAY [Configure the control machine (localhost)] ******************************
TASK [Gathering Facts] ********************************************************
[WARNING]: Host 'localhost' is using the discovered Python interpreter at '/usr/bin/python3.12', but futu
ok: [localhost]
TASK [Create a lab report directory] ******************************************
ok: [localhost]
TASK [Record the managed hosts] ***********************************************
ok: [localhost]
PLAY [Configure the managed node and deploy the app page] *********************
TASK [Gathering Facts] ********************************************************
ok: [lab3node]
TASK [Install nginx] **********************************************************
ok: [lab3node]
TASK [Deploy the application index page] **************************************
changed: [lab3node]
TASK [Start nginx (container has no systemd)] *********************************
ok: [lab3node]
TASK [Check nginx responds] ***************************************************
ok: [lab3node]
TASK [Show the deployed page] *************************************************
ok: [lab3node] => {
    "msg": "nginx responded HTTP 200 with the deployed page"
}
PLAY RECAP ********************************************************************
lab3node                   : ok=6    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
localhost                  : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
abhinav@LAPTOP-BSV7PCJK:~$ curl http://localhost:8081/   (deployed page on the managed node)
<h1>Lab 3 - configured by Ansible</h1>
<p>Host: lab3node</p>
<p>App version: 3.0.0</p>
### IDEMPOTENCY CHECK: second run should change nothing
abhinav@LAPTOP-BSV7PCJK:~$ ansible-playbook -i ansible/inventory.ini ansible/playbook.yml (2nd run)
[WARNING]: Host 'localhost' is using the discovered Python interpreter at '/usr/bin/python3.12', but futu
PLAY [Configure the control machine (localhost)] ******************************
ok: [localhost]
```

Fig: Playbook run: tasks applied on both targets

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
ok: [localhost]
ok: [localhost]
ok: [lab3node]
ok: [lab3node]
ok: [lab3node]
ok: [lab3node]
ok: [lab3node]
ok: [lab3node] => {
PLAY RECAP ********************************************************************
lab3node                   : ok=6    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
localhost                  : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
```

Fig: Deployed page and idempotent second run (changed=0)

Observations

- Two connection plugins in one inventory: local for the control machine, docker for the managed node.

- Idempotency: the second run reported changed=0 - Ansible only changes what differs from the declared state.
- The container has no systemd, so nginx was started with the init script (service nginx start).
- Port 8080 belongs to Jenkins, so the managed node's nginx is published on 8081.

Conclusion

All experiments were performed on Ubuntu 24.04 LTS with Ansible, Terraform, Prometheus, Grafana, the ELK stack and Selenium. The evidence (command logs, configuration files, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-3 under the docs/ directory.